ML Architecture

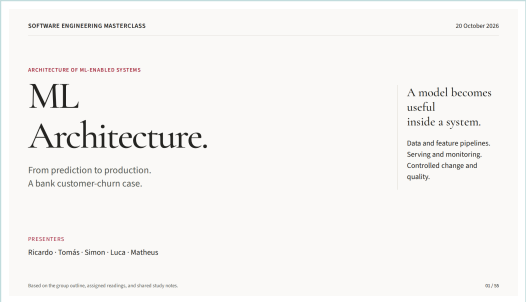
Speaker Notes

A bank customer-churn case — 20 October 2026

Each page shows one slide exactly as presenters will see it, the words to speak, and short delivery cues. The script is a speaking aid, not a transcript to read verbatim — adapt it to your own voice. Animated builds are marked; press the arrow key the stated number of times before moving to the next slide.

01 / 55 ML Architecture

Block 1 / Foundations • All presenters



CUES

- New block: **Block 1 / Foundations**.
- Presenter: **All presenters**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

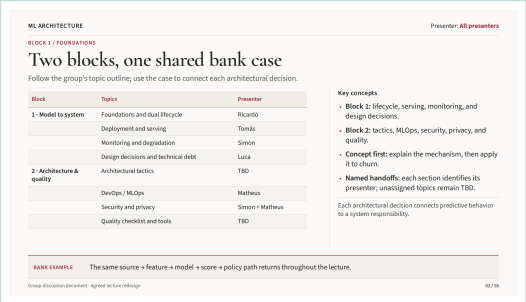
TO SAY

From a trained model to a maintainable, observable, governed system. **Software engineering** for systems that depend on learned behavior. **One shared case**: bank customer-churn prediction. **Two blocks**: model-to-system foundations, then architecture and quality. **Five presenters**: Ricardo, Tomás, Simon, Luca, and Matheus. Bank churn supplies the shared case. Examples, records, schedules, and incidents are illustrative. *A useful model needs an engineered system around it.*

Reading anchor: Bass, Ch. 14 · Kalinowski, Ch. 13–14

02 / 55 Two blocks, one shared bank case

Block 1 / Foundations • All presenters



CUES

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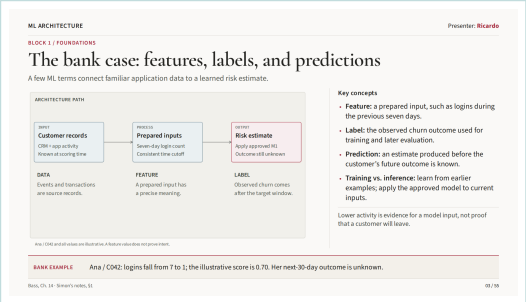
TO SAY

Follow the group's topic outline; use the case to connect each architectural decision. **Block 1**: lifecycle, serving, monitoring, and design decisions. **Block 2**: tactics, MLOps, security, privacy, and quality. **Concept first**: explain the mechanism, then apply it to churn. **Named handoffs**: each section identifies its presenter; unassigned topics remain TBD. *At our bank,* the same source → feature → model → score → policy path returns throughout the lecture. *Each architectural decision connects predictive behavior to a system responsibility.*

Reading anchor: Group discussion document · Agreed lecture redesign

03 / 55 The bank case: features, labels, and predictions

Block 1 / Foundations • Ricardo



CUES

- Presenter: **Ricardo**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
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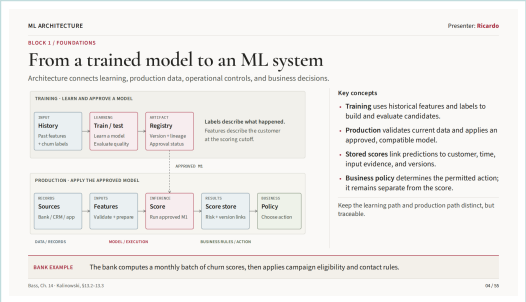
TO SAY

A few ML terms connect familiar application data to a learned risk estimate. **Feature**: a prepared input, such as logins during the previous seven days. **Label**: the observed churn outcome used for training and later evaluation. **Prediction**: an estimate produced before the customer's future outcome is known. **Training vs. inference**: learn from earlier examples; apply the approved model to current inputs. *At our bank,* ana / C042: logins fall from 7 to 1; the illustrative score is 0.70. Her next-30-day outcome is unknown. *Lower activity is evidence for a model input, not proof that a customer will leave.*

Reading anchor: Bass, Ch. 14 · Simon's notes, §1

04 / 55 From a trained model to an ML system

Block 1 / Foundations • Ricardo



CUES

- Presenter: **Ricardo**.
- Animated build: click 4 times to reveal the remaining steps before moving on.
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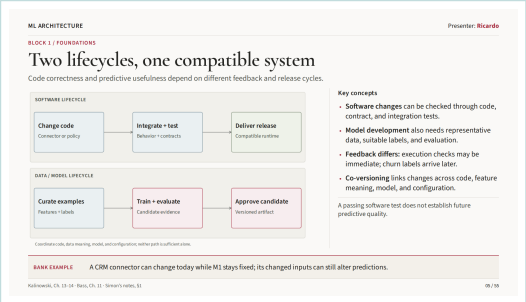
TO SAY

Architecture connects learning, production data, operational controls, and business decisions. **Training** uses historical features and labels to build and evaluate candidates. **Production** validates current data and applies an approved, compatible model. **Stored scores** link predictions to customer, time, input evidence, and versions. **Business policy** determines the permitted action; it remains separate from the score. *At our bank,* the bank computes a monthly batch of churn scores, then applies campaign eligibility and contact rules. *Keep the learning path and production path distinct, but traceable.*

Reading anchor: Bass, Ch. 14 · Kalinowski, §13.2–13.3

05 / 55 Two lifecycles, one compatible system

Block 1 / Foundations • Ricardo



CUES

- Presenter: **Ricardo**.
- Animated build: click 3 times to reveal the remaining steps before moving on.
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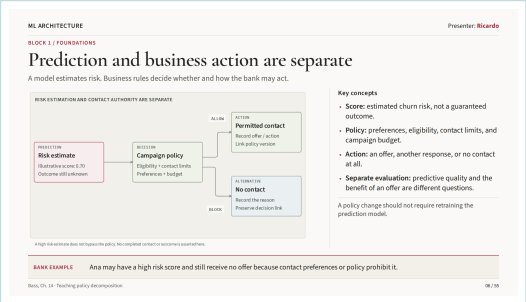
TO SAY

Code correctness and predictive usefulness depend on different feedback and release cycles. **Software changes** can be checked through code, contract, and integration tests. **Model development** also needs representative data, suitable labels, and evaluation. **Feedback differs**: execution checks may be immediate; churn labels arrive later. **Co-versioning** links changes across code, feature meaning, model, and configuration. *At our bank,* a CRM connector can change today while M1 stays fixed; its changed inputs can still alter predictions. *A passing software test does not establish future predictive quality.*

Reading anchor: Kalinowski, Ch. 13–14 · Bass, Ch. 11 · Simon’s notes, §1

06 / 55 Prediction and business action are separate

Block 1 / Foundations • Ricardo



CUES

- Presenter: **Ricardo**.
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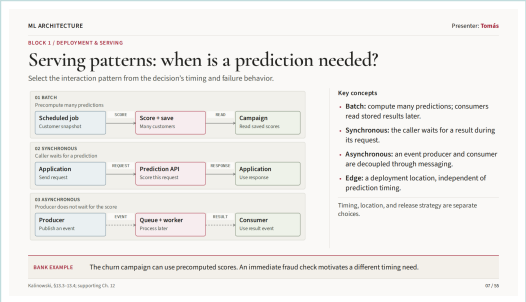
TO SAY

A model estimates risk. Business rules decide whether and how the bank may act. **Score**: estimated churn risk, not a guaranteed outcome. **Policy**: preferences, eligibility, contact limits, and campaign budget. **Action**: an offer, another response, or no contact at all. **Separate evaluation**: predictive quality and the benefit of an offer are different questions. *At our bank,* ana may have a high risk score and still receive no offer because contact preferences or policy prohibit it. *A policy change should not require retraining the prediction model.*

Reading anchor: Bass, Ch. 14 · Teaching policy decomposition

07 / 55 Serving patterns: when is a prediction needed?

Block 1 / Deployment & serving • Tomás



CUES

- New block: **Block 1 / Deployment & serving**.
- Presenter: **Tomás**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
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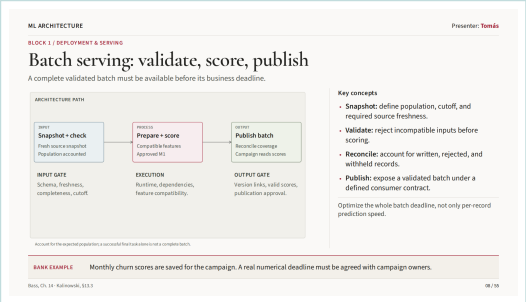
TO SAY

Select the interaction pattern from the decision's timing and failure behavior. **Batch**: compute many predictions; consumers read stored results later. **Synchronous**: the caller waits for a result during its request. **Asynchronous**: an event producer and consumer are decoupled through messaging. **Edge**: a deployment location, independent of prediction timing. *At our bank,* the churn campaign can use precomputed scores. An immediate fraud check motivates a different timing need. *Timing, location, and release strategy are separate choices.*

Reading anchor: Kalinowski, §13.3–13.4; supporting Ch. 12

08 / 55 Batch serving: validate, score, publish

Block 1 / Deployment & serving • Tomás



CUES

- Presenter: **Tomás**.
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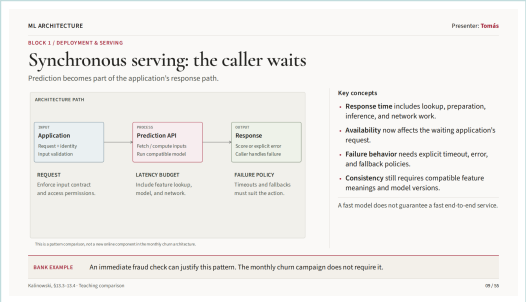
TO SAY

A complete validated batch must be available before its business deadline. **Snapshot**: define population, cutoff, and required source freshness. **Validate**: reject incompatible inputs before scoring. **Reconcile**: account for written, rejected, and withheld records. **Publish**: expose a validated batch under a defined consumer contract. *At our bank,* monthly churn scores are saved for the campaign. A real numerical deadline must be agreed with campaign owners. *Optimize the whole batch deadline, not only per-record prediction speed.*

Reading anchor: Bass, Ch. 14 · Kalinowski, §13.3

09 / 55 Synchronous serving: the caller waits

Block 1 / Deployment & serving • Tomás



CUES

- Presenter: **Tomás**.
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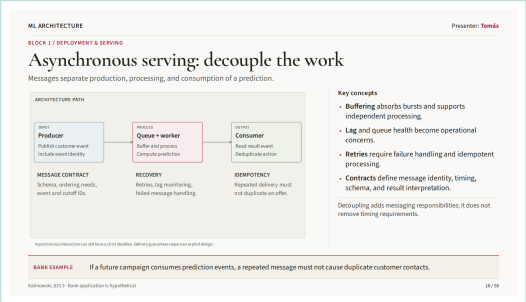
TO SAY

Prediction becomes part of the application's response path. **Response time** includes lookup, preparation, inference, and network work. **Availability** now affects the waiting application's request. **Failure behavior** needs explicit timeout, error, and fallback policies. **Consistency** still requires compatible feature meanings and model versions. *At our bank,* an immediate fraud check can justify this pattern. The monthly churn campaign does not require it. *A fast model does not guarantee a fast end-to-end service.*

Reading anchor: Kalinowski, §13.3–13.4 · Teaching comparison

10 / 55 Asynchronous serving: decouple the work

Block 1 / Deployment & serving • Tomás



CUES

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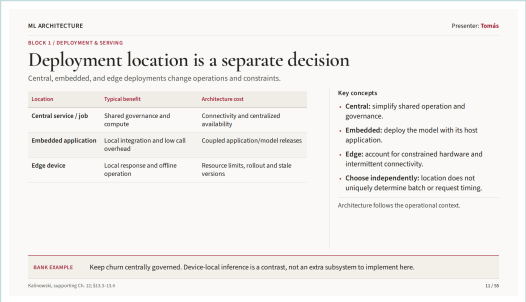
TO SAY

Messages separate production, processing, and consumption of a prediction. **Buffering** absorbs bursts and supports independent processing. **Lag** and queue health become operational concerns. **Retries** require failure handling and idempotent processing. **Contracts** define message identity, timing, schema, and result interpretation. *At our bank,* if a future campaign consumes prediction events, a repeated message must not cause duplicate customer contacts. *Decoupling adds messaging responsibilities; it does not remove timing requirements.*

Reading anchor: Kalinowski, §13.3 · Bank application is hypothetical

11 / 55 Deployment location is a separate decision

Block 1 / Deployment & serving • Tomás



CUES

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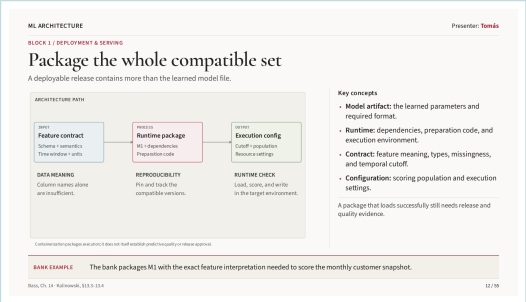
TO SAY

Central, embedded, and edge deployments change operations and constraints. **Central**: simplify shared operation and governance. **Embedded**: deploy the model with its host application. **Edge**: account for constrained hardware and intermittent connectivity. **Choose independently**: location does not uniquely determine batch or request timing. *At our bank,* keep churn centrally governed. Device-local inference is a contrast, not an extra subsystem to implement here. *Architecture follows the operational context.*

Reading anchor: Kalinowski, supporting Ch. 12; §13.3–13.4

12 / 55 Package the whole compatible set

Block 1 / Deployment & serving • Tomás



CUES

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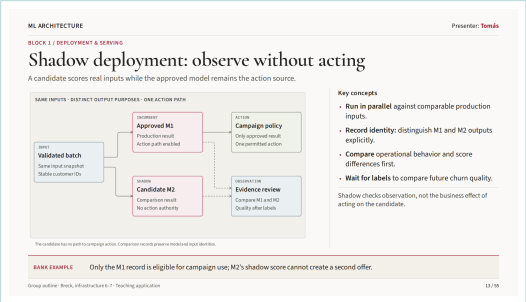
TO SAY

A deployable release contains more than the learned model file. **Model artifact**: the learned parameters and required format. **Runtime**: dependencies, preparation code, and execution environment. **Contract**: feature meaning, types, missingness, and temporal cutoff. **Configuration**: scoring population and execution settings. *At our bank,* the bank packages M1 with the exact feature interpretation needed to score the monthly customer snapshot. *A package that loads successfully still needs release and quality evidence.*

Reading anchor: Bass, Ch. 14 · Kalinowski, §13.3–13.4

13 / 55 Shadow deployment: observe without acting

Block 1 / Deployment & serving • Tomás



CUES

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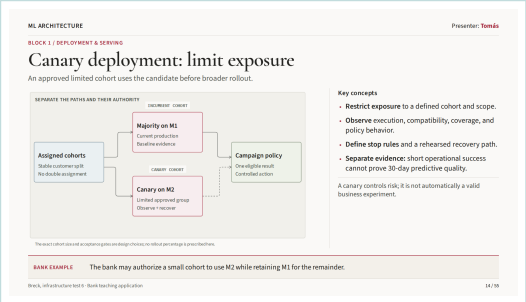
TO SAY

A candidate scores real inputs while the approved model remains the action source. **Run in parallel** against comparable production inputs. **Record identity**: distinguish M1 and M2 outputs explicitly. **Compare** operational behavior and score differences first. **Wait for labels** to compare future churn quality. *At our bank,* only the M1 record is eligible for campaign use; M2's shadow score cannot create a second offer. *Shadow checks observation, not the business effect of acting on the candidate.*

Reading anchor: Group outline · Breck, infrastructure 6–7 · Teaching application

14 / 55 Canary deployment: limit exposure

Block 1 / Deployment & serving • Tomás



CUES

- Presenter: **Tomás**.
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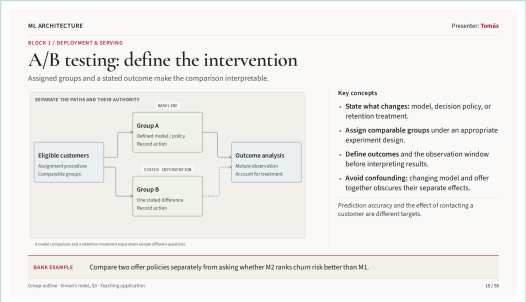
TO SAY

An approved limited cohort uses the candidate before broader rollout. **Restrict exposure** to a defined cohort and scope. **Observe** execution, compatibility, coverage, and policy behavior. **Define stop rules** and a rehearsed recovery path. **Separate evidence**: short operational success cannot prove 30-day predictive quality. *At our bank,* the bank may authorize a small cohort to use M2 while retaining M1 for the remainder. *A canary controls risk; it is not automatically a valid business experiment.*

Reading anchor: Breck, infrastructure test 6 · Bank teaching application

15 / 55 A/B testing: define the intervention

Block 1 / Deployment & serving • Tomás



CUES

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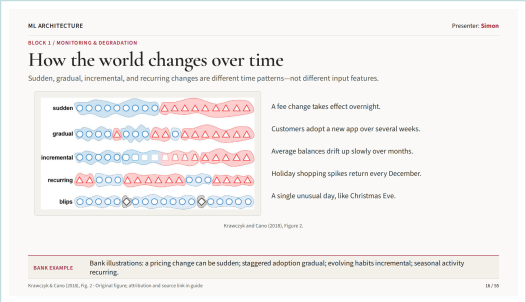
TO SAY

Assigned groups and a stated outcome make the comparison interpretable. **State what changes**: model, decision policy, or retention treatment. **Assign comparable groups** under an appropriate experiment design. **Define outcomes** and the observation window before interpreting results. **Avoid confounding**: changing model and offer together obscures their separate effects. *At our bank,* compare two offer policies separately from asking whether M2 ranks churn risk better than M1. *Prediction accuracy and the effect of contacting a customer are different targets.*

Reading anchor: Group outline · Simon's notes, §3 · Teaching application

16 / 55 How the world changes over time

Block 1 / Monitoring & degradation • Simon



CUES

- New block: **Block 1 / Monitoring & degradation**.
- Presenter: **Simon**.
- Pause briefly before the bank example — let the concept land first.
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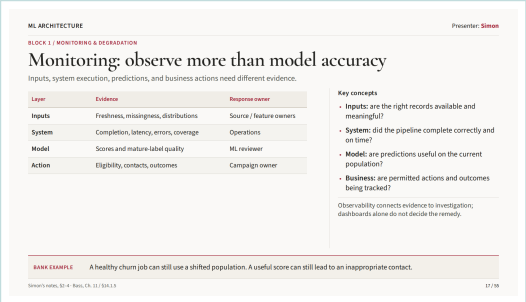
TO SAY

Sudden, gradual, incremental, and recurring changes are different time patterns—not different input features. **Sudden**: the old regime switches abruptly. **Gradual**: old and new regimes coexist during a transition. **Incremental / recurring**: intermediate change differs from a returning regime. **Blips**: isolated noise is not necessarily a lasting regime change. Bank illustrations: a pricing change can be sudden; staggered adoption gradual; evolving habits incremental; seasonal activity recurring. *First ask how change unfolds over time. Then ask what changed: input data, outcome relationships, or data validity.*

Reading anchor: Krawczyk & Cano (2018), Fig. 2 · Original figure; attribution and source link in guide

17 / 55 Monitoring: observe more than model accuracy

Block 1 / Monitoring & degradation • Simon



CUES

- Presenter: **Simon**.
- Pause briefly before the bank example — let the concept land first.
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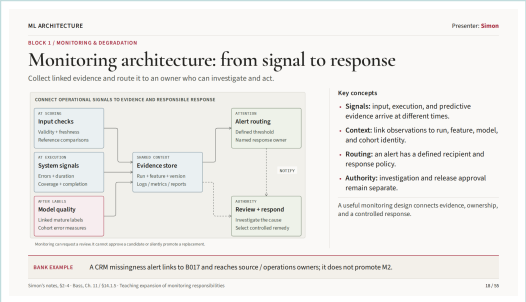
TO SAY

Inputs, system execution, predictions, and business actions need different evidence. **Inputs**: are the right records available and meaningful? **System**: did the pipeline complete correctly and on time? **Model**: are predictions useful on the current population? **Business**: are permitted actions and outcomes being tracked? *At our bank,* a healthy churn job can still use a shifted population. A useful score can still lead to an inappropriate contact. *Observability connects evidence to investigation; dashboards alone do not decide the remedy.*

Reading anchor: Simon’s notes, §2–4 · Bass, Ch. 11 / §14.1.5

18 / 55 Monitoring architecture: from signal to response

Block 1 / Monitoring & degradation • Simon



CUES

- Presenter: **Simon**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
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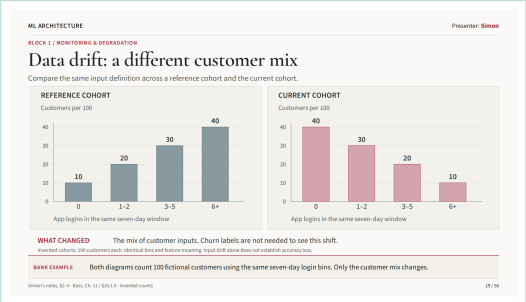
TO SAY

Collect linked evidence and route it to an owner who can investigate and act. **Signals**: input, execution, and predictive evidence arrive at different times. **Context**: link observations to run, feature, model, and cohort identity. **Routing**: an alert has a defined recipient and response policy. **Authority**: investigation and release approval remain separate. *At our bank,* a CRM missingness alert links to B017 and reaches source / operations owners; it does not promote M2. *A useful monitoring design connects evidence, ownership, and a controlled response.*

Reading anchor: Simon’s notes, §2–4 · Bass, Ch. 11 / §14.1.5 · Teaching expansion of monitoring responsibilities

19 / 55 Data drift: a different customer mix

Block 1 / Monitoring & degradation • Simon



CUES

- Presenter: **Simon**.
- Animated build: click 1 time to reveal the remaining steps before moving on.
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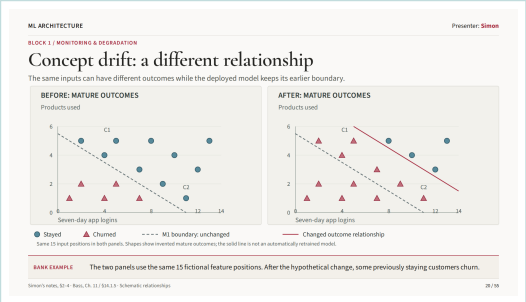
TO SAY

Compare the same input definition across a reference cohort and the current cohort. **What changes**: the distribution of valid model inputs. **Bank example**: a campaign attracts more low-activity customers. **Available now**: compare inputs without waiting for churn labels. **Limit**: a changed distribution alone does not prove lost accuracy. *At our bank,* both diagrams count 100 fictional customers using the same seven-day login bins. Only the customer mix changes. *Choose a meaningful reference, comparable cohorts, and a defined response.*

Reading anchor: Simon’s notes, §2–4 · Bass, Ch. 11 / §14.1.5 · Invented counts

20 / 55 Concept drift: a different relationship

Block 1 / Monitoring & degradation • Simon



CUES

- Presenter: **Simon**.
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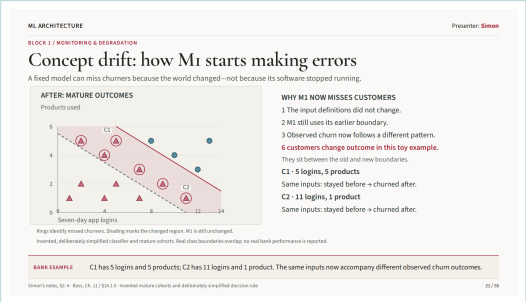
TO SAY

The same inputs can have different outcomes while the deployed model keeps its earlier boundary. **What changes**: churn risk given the same valid input information. **Bank example**: new fees make active customers leave more often. **Evidence**: link predictions to mature, reliable churn outcomes. **Model status**: M1 stays fixed until a separately evaluated release changes it. *At our bank,* the two panels use the same 15 fictional feature positions. After the hypothetical change, some previously staying customers churn. *A missing field is not concept drift; a statistical alarm is not a retraining approval.*

Reading anchor: Simon’s notes, §2–4 · Bass, Ch. 11 / §14.1.5 · Schematic relationships

21 / 55 Concept drift: how M1 starts making errors

Block 1 / Monitoring & degradation • Simon



CUES

- Presenter: **Simon**.
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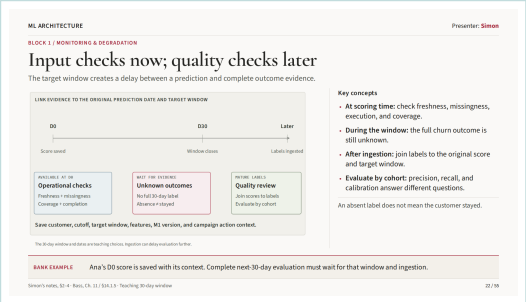
TO SAY

A fixed model can miss churners because the world changed—not because its software stopped running. **Inputs stay fixed**: the two cohorts use identical feature positions. **M1 stays fixed**: its earlier decision boundary is unchanged. **Outcomes change**: a different relationship creates missed churners. **Evidence**: the example uses mature labels, not outcomes already known on scoring day. *At our bank,* c1 has 5 logins and 5 products; C2 has 11 logins and 1 product. The same inputs now accompany different observed churn outcomes. *Changed labels reveal the mismatch. Investigate causes, evaluate candidates, and use controlled release rather than automatic promotion.*

Reading anchor: Simon's notes, §2–4 · Bass, Ch. 11 / §14.1.5 · Invented mature cohorts and deliberately simplified decision rule

22 / 55 Input checks now; quality checks later

Block 1 / Monitoring & degradation • Simon



CUES

- Presenter: **Simon**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
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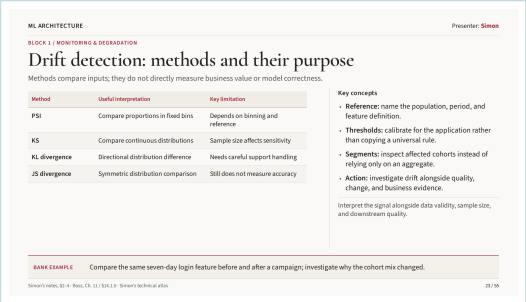
TO SAY

The target window creates a delay between a prediction and complete outcome evidence. **At scoring time**: check freshness, missingness, execution, and coverage. **During the window**: the full churn outcome is still unknown. **After ingestion**: join labels to the original score and target window. **Evaluate by cohort**: precision, recall, and calibration answer different questions. *At our bank,* ana's D0 score is saved with its context. Complete next-30-day evaluation must wait for that window and ingestion. *An absent label does not mean the customer stayed.*

Reading anchor: Simon's notes, §2–4 · Bass, Ch. 11 / §14.1.5 · Teaching 30-day window

23 / 55 Drift detection: methods and their purpose

Block 1 / Monitoring & degradation • Simon



CUES

- Presenter: **Simon**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

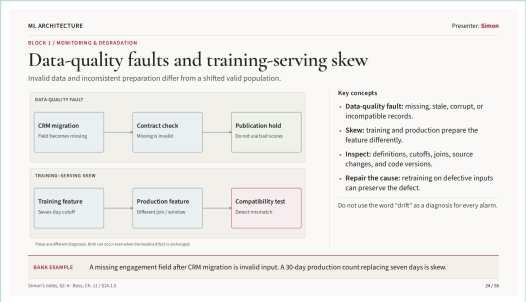
TO SAY

Methods compare inputs; they do not directly measure business value or model correctness. **Reference**: name the population, period, and feature definition. **Thresholds**: calibrate for the application rather than copying a universal rule. **Segments**: inspect affected cohorts instead of relying only on an aggregate. **Action**: investigate drift alongside quality, change, and business evidence. *At our bank,* compare the same seven-day login feature before and after a campaign; investigate why the cohort mix changed. *Interpret the signal alongside data validity, sample size, and downstream quality.*

Reading anchor: Simon’s notes, §2–4 · Bass, Ch. 11 / §14.1.5 · Simon’s technical atlas

24 / 55 Data-quality faults and training-serving skew

Block 1 / Monitoring & degradation • Simon



CUES

- Presenter: **Simon**.
- Animated build: click 3 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

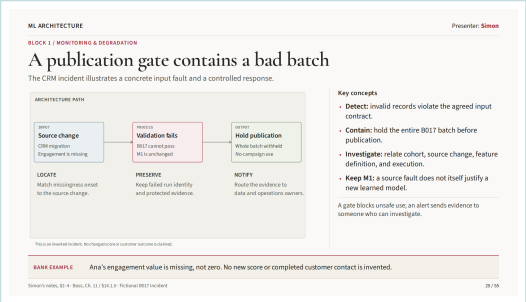
TO SAY

Invalid data and inconsistent preparation differ from a shifted valid population. **Data-quality fault**: missing, stale, corrupt, or incompatible records. **Skew**: training and production prepare the feature differently. **Inspect**: definitions, cutoffs, joins, source changes, and code versions. **Repair the cause**: retraining on defective inputs can preserve the defect. *At our bank,* a missing engagement field after CRM migration is invalid input. A 30-day production count replacing seven days is skew. *Do not use the word "drift" as a diagnosis for every alarm.*

Reading anchor: Simon's notes, §2–4 · Bass, Ch. 11 / §14.1.5

25 / 55 A publication gate contains a bad batch

Block 1 / Monitoring & degradation • Simon



CUES

- Presenter: **Simon**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

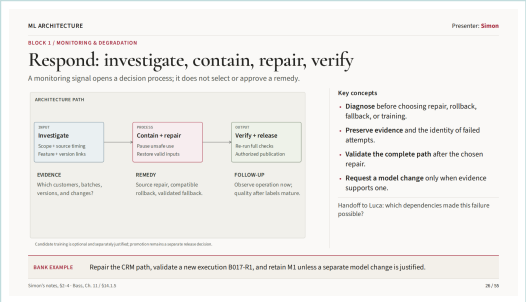
TO SAY

The CRM incident illustrates a concrete input fault and a controlled response. **Detect**: invalid records violate the agreed input contract. **Contain**: hold the entire B017 batch before publication. **Investigate**: relate cohort, source change, feature definition, and execution. **Keep M1**: a source fault does not itself justify a new learned model. *At our bank,* ana's engagement value is missing, not zero. No new score or completed customer contact is invented. *A gate blocks unsafe use; an alert sends evidence to someone who can investigate.*

Reading anchor: Simon's notes, §2–4 · Bass, Ch. 11 / §14.1.5 · Fictional B017 incident

26 / 55 Respond: investigate, contain, repair, verify

Block 1 / Monitoring & degradation • Simon



CUES

- Presenter: **Simon**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

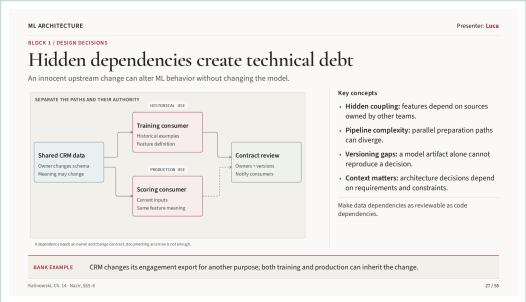
TO SAY

A monitoring signal opens a decision process; it does not select or approve a remedy. **Diagnose** before choosing repair, rollback, fallback, or training. **Preserve evidence** and the identity of failed attempts. **Validate the complete path** after the chosen repair. **Request a model change** only when evidence supports one. *At our bank,* repair the CRM path, validate a new execution B017-R1, and retain M1 unless a separate model change is justified. *Handoff to Luca: which dependencies made this failure possible?*

Reading anchor: Simon’s notes, §2–4 · Bass, Ch. 11 / §14.1.5

27 / 55 Hidden dependencies create technical debt

Block 1 / Design decisions • Luca



CUES

- New block: **Block 1 / Design decisions**.
- Presenter: **Luca**.
- Animated build: click 3 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

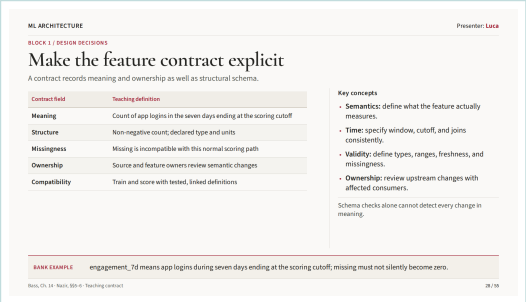
TO SAY

An innocent upstream change can alter ML behavior without changing the model. **Hidden coupling**: features depend on sources owned by other teams. **Pipeline complexity**: parallel preparation paths can diverge. **Versioning gaps**: a model artifact alone cannot reproduce a decision. **Context matters**: architecture decisions depend on requirements and constraints. *At our bank,* cRM changes its engagement export for another purpose; both training and production can inherit the change. *Make data dependencies as reviewable as code dependencies.*

Reading anchor: Kalinowski, Ch. 14 · Nazir, §§5–6

28 / 55 Make the feature contract explicit

Block 1 / Design decisions • Luca



CUES

- Presenter: **Luca**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

A contract records meaning and ownership as well as structural schema. **Semantics**: define what the feature actually measures. **Time**: specify window, cutoff, and joins consistently. **Validity**: define types, ranges, freshness, and missingness. **Ownership**: review upstream changes with affected consumers. *At our bank,* engagement_7d means app logins during seven days ending at the scoring cutoff; missing must not silently become zero. *Schema checks alone cannot detect every change in meaning.*

Reading anchor: Bass, Ch. 14 · Nazir, §§5–6 · Teaching contract

29 / 55 Separate responsibilities where change happens

Block 1 / Design decisions • Luca



CUES

- Presenter: **Luca**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

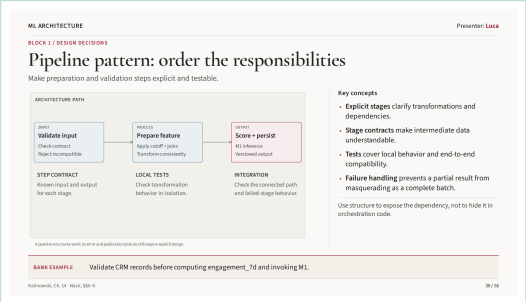
TO SAY

Clear interfaces let different concerns evolve without unnecessary relearning. **Single responsibility**: identify each module's reason to change. **Interfaces**: specify data and behavior at each boundary. **Dependency injection**: provide implementations through explicit contracts. **Deployment**: logical modules do not automatically require separate services. *At our bank,* a connector fix changes preparation; a contact-frequency rule changes campaign policy; a new M2 changes inference. *Separate concerns first; choose deployment units from the actual trade-offs.*

Reading anchor: Kalinowski, Ch. 14 · Nazir, §§5–6

30 / 55 Pipeline pattern: order the responsibilities

Block 1 / Design decisions • Luca



CUES

- Presenter: **Luca**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

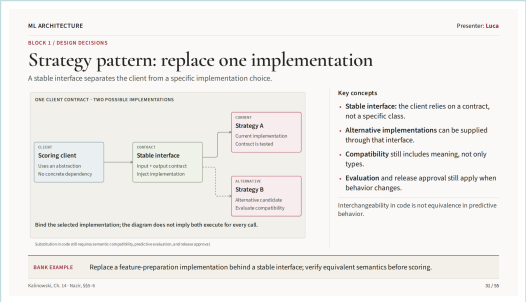
TO SAY

Make preparation and validation steps explicit and testable. **Explicit stages** clarify transformations and dependencies. **Stage contracts** make intermediate data understandable. **Tests** cover local behavior and end-to-end compatibility. **Failure handling** prevents a partial result from masquerading as a complete batch. *At our bank,* validate CRM records before computing engagement_7d and invoking M1. *Use structure to expose the dependency, not to hide it in orchestration code.*

Reading anchor: Kalinowski, Ch. 14 · Nazir, §§5–6

31 / 55 Strategy pattern: replace one implementation

Block 1 / Design decisions • Luca



CUES

- Presenter: **Luca**.
- Animated build: click 3 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

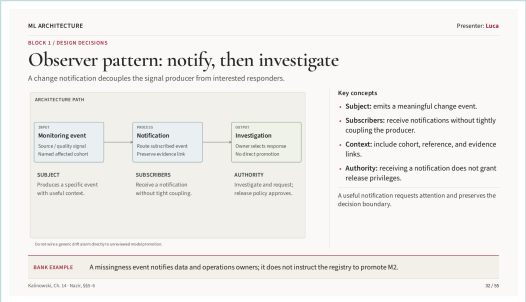
TO SAY

A stable interface separates the client from a specific implementation choice. **Stable interface**: the client relies on a contract, not a specific class. **Alternative implementations** can be supplied through that interface. **Compatibility** still includes meaning, not only types. **Evaluation** and release approval still apply when behavior changes. *At our bank,* replace a feature-preparation implementation behind a stable interface; verify equivalent semantics before scoring. *Interchangeability in code is not equivalence in predictive behavior.*

Reading anchor: Kalinowski, Ch. 14 · Nazir, §§5–6

32 / 55 Observer pattern: notify, then investigate

Block 1 / Design decisions • Luca



CUES

- Presenter: **Luca**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

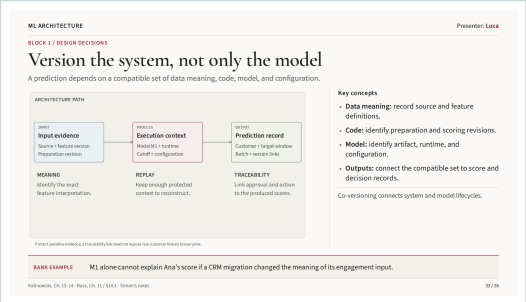
TO SAY

A change notification decouples the signal producer from interested responders. **Subject**: emits a meaningful change event. **Subscribers**: receive notifications without tightly coupling the producer. **Context**: include cohort, reference, and evidence links. **Authority**: receiving a notification does not grant release privileges. *At our bank,* a missingness event notifies data and operations owners; it does not instruct the registry to promote M2. *A useful notification requests attention and preserves the decision boundary.*

Reading anchor: Kalinowski, Ch. 14 · Nazir, §§5–6

33 / 55 Version the system, not only the model

Block 1 / Design decisions • Luca



CUES

- Presenter: **Luca**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

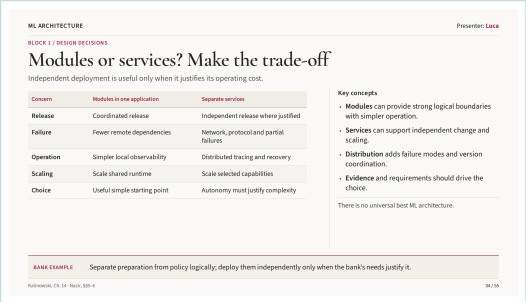
TO SAY

A prediction depends on a compatible set of data meaning, code, model, and configuration. **Data meaning**: record source and feature definitions. **Code**: identify preparation and scoring revisions. **Model**: identify artifact, runtime, and configuration. **Outputs**: connect the compatible set to score and decision records. *At our bank,* m1 alone cannot explain Ana's score if a CRM migration changed the meaning of its engagement input. *Co-versioning connects system and model lifecycles.*

Reading anchor: Kalinowski, Ch. 13–14 · Bass, Ch. 11 / §14.1 · Simon's notes

34 / 55 Modules or services? Make the trade-off

Block 1 / Design decisions • Luca



CUES

- Presenter: **Luca**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

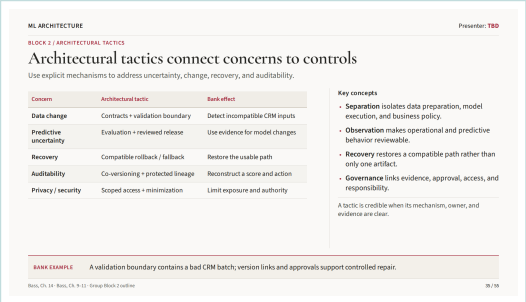
TO SAY

Independent deployment is useful only when it justifies its operating cost. **Modules** can provide strong logical boundaries with simpler operation. **Services** can support independent change and scaling. **Distribution** adds failure modes and version coordination. **Evidence** and requirements should drive the choice. *At our bank,* separate preparation from policy logically; deploy them independently only when the bank's needs justify it. *There is no universal best ML architecture.*

Reading anchor: Kalinowski, Ch. 14 · Nazir, §§5–6

35 / 55 Architectural tactics connect concerns to controls

Block 2 / Architectural tactics • TBD



CUES

- New block: **Block 2 / Architectural tactics**.
- Presenter: **TBD**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

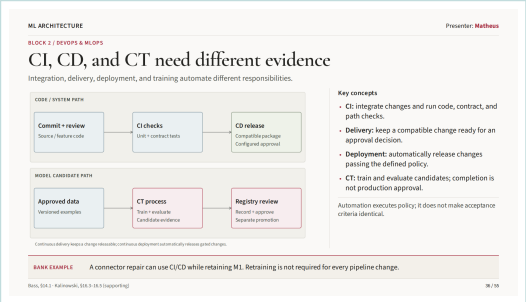
TO SAY

Use explicit mechanisms to address uncertainty, change, recovery, and auditability. **Separation** isolates data preparation, model execution, and business policy. **Observation** makes operational and predictive behavior reviewable. **Recovery** restores a compatible path rather than only one artifact. **Governance** links evidence, approval, access, and responsibility. *At our bank,* a validation boundary contains a bad CRM batch; version links and approvals support controlled repair. *A tactic is credible when its mechanism, owner, and evidence are clear.*

Reading anchor: Bass, Ch. 14 · Bass, Ch. 9–11 · Group Block 2 outline

36 / 55 CI, CD, and CT need different evidence

Block 2 / DevOps & MLOps • Matheus



CUES

- New block: **Block 2 / DevOps & MLOps**.
- Presenter: **Matheus**.
- Animated build: click 3 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

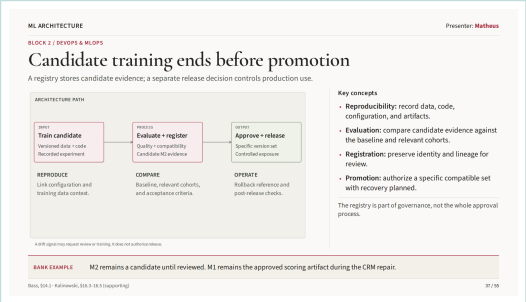
TO SAY

Integration, delivery, deployment, and training automate different responsibilities. **CI**: integrate changes and run code, contract, and path checks. **Delivery**: keep a compatible change ready for an approval decision. **Deployment**: automatically release changes passing the defined policy. **CT**: train and evaluate candidates; completion is not production approval. *At our bank,* a connector repair can use CI/CD while retaining M1. Retraining is not required for every pipeline change. *Automation executes policy; it does not make acceptance criteria identical.*

Reading anchor: Bass, §14.1 · Kalinowski, §16.3–16.5 (supporting)

37 / 55 Candidate training ends before promotion

Block 2 / DevOps & MLOps • Matheus



CUES

- Presenter: **Matheus**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

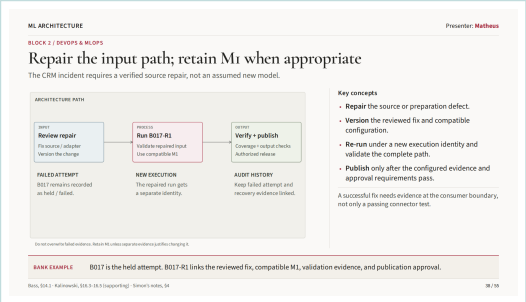
TO SAY

A registry stores candidate evidence; a separate release decision controls production use. **Reproducibility**: record data, code, configuration, and artifacts. **Evaluation**: compare candidate evidence against the baseline and relevant cohorts. **Registration**: preserve identity and lineage for review. **Promotion**: authorize a specific compatible set with recovery planned. *At our bank,* m2 remains a candidate until reviewed. M1 remains the approved scoring artifact during the CRM repair. *The registry is part of governance, not the whole approval process.*

Reading anchor: Bass, §14.1 · Kalinowski, §16.3–16.5 (supporting)

38 / 55 Repair the input path; retain M1 when appropriate

Block 2 / DevOps & MLOps • Matheus



CUES

- Presenter: **Matheus**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
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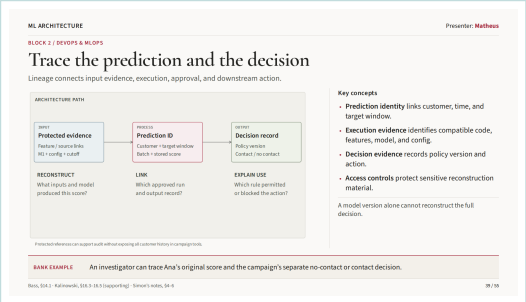
TO SAY

The CRM incident requires a verified source repair, not an assumed new model. **Repair** the source or preparation defect. **Version** the reviewed fix and compatible configuration. **Re-run** under a new execution identity and validate the complete path. **Publish** only after the configured evidence and approval requirements pass. *At our bank,* b017 is the held attempt. B017-R1 links the reviewed fix, compatible M1, validation evidence, and publication approval. *A successful fix needs evidence at the consumer boundary, not only a passing connector test.*

Reading anchor: Bass, §14.1 · Kalinowski, §16.3–16.5 (supporting) · Simon’s notes, §4

39 / 55 Trace the prediction and the decision

Block 2 / DevOps & MLOps • Matheus



CUES

- Presenter: **Matheus**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

Lineage connects input evidence, execution, approval, and downstream action. **Prediction identity** links customer, time, and target window. **Execution evidence** identifies compatible code, features, model, and config. **Decision evidence** records policy version and action. **Access controls** protect sensitive reconstruction material. *At our bank,* an investigator can trace Ana's original score and the campaign's separate no-contact or contact decision. *A model version alone cannot reconstruct the full decision.*

Reading anchor: Bass, §14.1 · Kalinowski, §16.3–16.5 (supporting) · Simon's notes, §4–6

40 / 55 Recovery restores a compatible path

Block 2 / DevOps & MLOps • Matheus



CUES

- Presenter: **Matheus**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

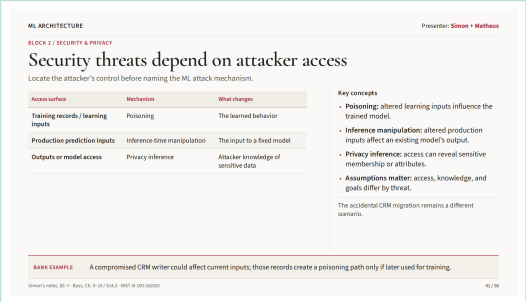
TO SAY

Rehearse restoration before release, and verify it at the consumer boundary. **Know the target**: identify the last compatible release or validated fallback. **Rehearse**: test the recovery procedure before relying on it. **Restore the set**: include data meaning, runtime, and configuration. **Verify**: inspect stored scores and actual consumer behavior. *At our bank,* rolling M1 back cannot repair a CRM field that is still missing; the input path must be restored. *Handoff to security: who may read, change, approve, or operate these paths?*

Reading anchor: Breck, infrastructure tests 6–7 · Simon’s notes, §4

41 / 55 Security threats depend on attacker access

Block 2 / Security & privacy • Simon + Matheus



CUES

- New block: **Block 2 / Security & privacy**.
- Presenter: **Simon + Matheus**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

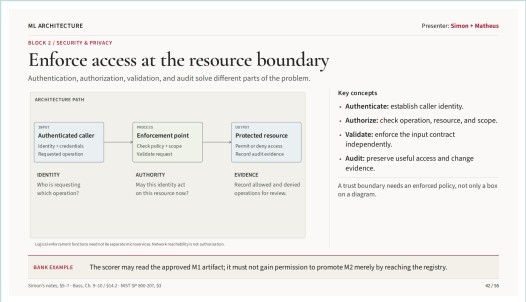
TO SAY

Locate the attacker's control before naming the ML attack mechanism. **Poisoning**: altered learning inputs influence the trained model. **Inference manipulation**: altered production inputs affect an existing model's output. **Privacy inference**: access can reveal sensitive membership or attributes. **Assumptions matter**: access, knowledge, and goals differ by threat. *At our bank,* a compromised CRM writer could affect current inputs; those records create a poisoning path only if later used for training. *The accidental CRM migration remains a different scenario.*

Reading anchor: Simon's notes, §5–7 · Bass, Ch. 9–10 / §14.2 · NIST AI 100-2e2025

42 / 55 Enforce access at the resource boundary

Block 2 / Security & privacy • Simon + Matheus



CUES

- Presenter: **Simon + Matheus**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

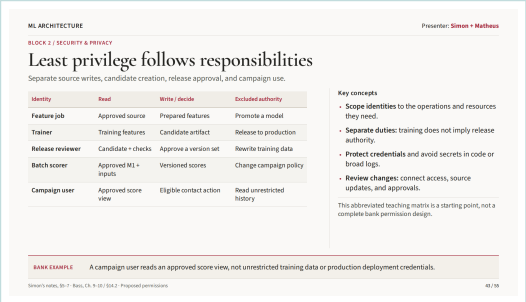
TO SAY

Authentication, authorization, validation, and audit solve different parts of the problem. **Authenticate**: establish caller identity. **Authorize**: check operation, resource, and scope. **Validate**: enforce the input contract independently. **Audit**: preserve useful access and change evidence. *At our bank,* the scorer may read the approved M1 artifact; it must not gain permission to promote M2 merely by reaching the registry. *A trust boundary needs an enforced policy, not only a box on a diagram.*

Reading anchor: Simon's notes, §5–7 · Bass, Ch. 9–10 / §14.2 · NIST SP 800-207, §3

43 / 55 Least privilege follows responsibilities

Block 2 / Security & privacy • Simon + Matheus



CUES

- Presenter: **Simon + Matheus**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

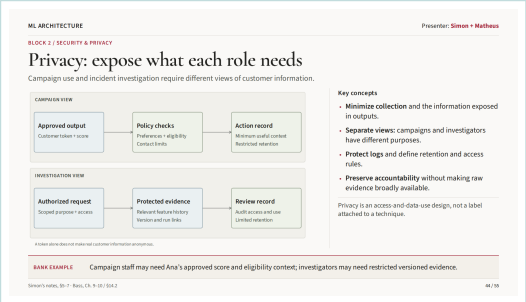
TO SAY

Separate source writes, candidate creation, release approval, and campaign use. **Scope identities** to the operations and resources they need. **Separate duties**: training does not imply release authority. **Protect credentials** and avoid secrets in code or broad logs. **Review changes**: connect access, source updates, and approvals. *At our bank,* a campaign user reads an approved score view, not unrestricted training data or production deployment credentials. *This abbreviated teaching matrix is a starting point, not a complete bank permission design.*

Reading anchor: Simon's notes, §5–7 · Bass, Ch. 9–10 / §14.2 · Proposed permissions

44 / 55 Privacy: expose what each role needs

Block 2 / Security & privacy • Simon + Matheus



CUES

- Presenter: **Simon + Matheus**.
- Animated build: click 3 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

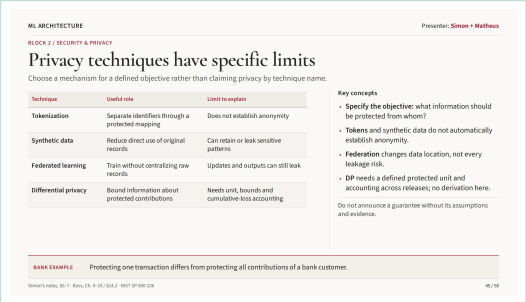
TO SAY

Campaign use and incident investigation require different views of customer information. **Minimize collection** and the information exposed in outputs. **Separate views**: campaigns and investigators have different purposes. **Protect logs** and define retention and access rules. **Preserve accountability** without making raw evidence broadly available. *At our bank,* campaign staff may need Ana's approved score and eligibility context; investigators may need restricted versioned evidence. *Privacy is an access-and-data-use design, not a label attached to a technique.*

Reading anchor: Simon's notes, §5–7 · Bass, Ch. 9–10 / §14.2

45 / 55 Privacy techniques have specific limits

Block 2 / Security & privacy • Simon + Matheus



CUES

- Presenter: **Simon + Matheus**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

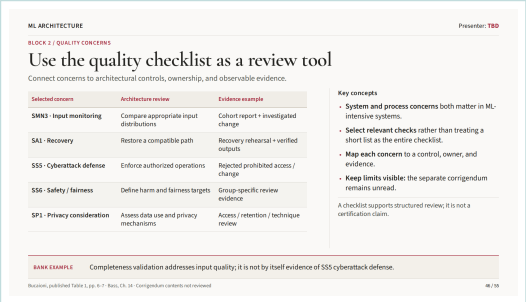
TO SAY

Choose a mechanism for a defined objective rather than claiming privacy by technique name. **Specify the objective**: what information should be protected from whom? **Tokens** and synthetic data do not automatically establish anonymity. **Federation** changes data location, not every leakage risk. **DP** needs a defined protected unit and accounting across releases; no derivation here. *At our bank,* protecting one transaction differs from protecting all contributions of a bank customer. *Do not announce a guarantee without its assumptions and evidence.*

Reading anchor: Simon's notes, §5–7 · Bass, Ch. 9–10 / §14.2 · NIST SP 800-226

46 / 55 Use the quality checklist as a review tool

Block 2 / Quality concerns • TBD



CUES

- New block: **Block 2 / Quality concerns**.
- Presenter: **TBD**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

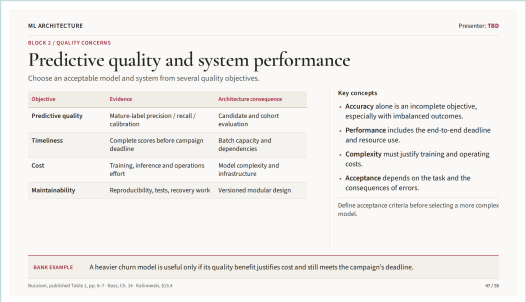
TO SAY

Connect concerns to architectural controls, ownership, and observable evidence. **System and process concerns** both matter in ML-intensive systems. **Select relevant checks** rather than treating a short list as the entire checklist. **Map each concern** to a control, owner, and evidence. **Keep limits visible**: the separate corrigendum remains unread. *At our bank,* completeness validation addresses input quality; it is not by itself evidence of SS5 cyberattack defense. *A checklist supports structured review; it is not a certification claim.*

Reading anchor: Bucaioni, published Table 1, pp. 6–7 · Bass, Ch. 14 · Corrigendum contents not reviewed

47 / 55 Predictive quality and system performance

Block 2 / Quality concerns • TBD



CUES

- Presenter: **TBD**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

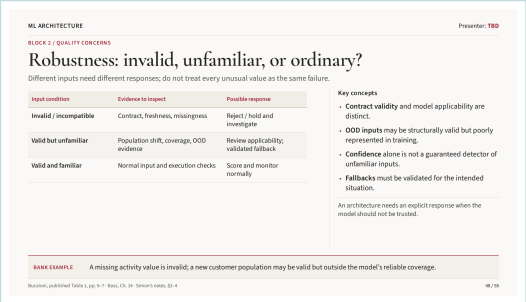
TO SAY

Choose an acceptable model and system from several quality objectives. **Accuracy** alone is an incomplete objective, especially with imbalanced outcomes. **Performance** includes the end-to-end deadline and resource use. **Complexity** must justify training and operating costs. **Acceptance** depends on the task and the consequences of errors. *At our bank,* a heavier churn model is useful only if its quality benefit justifies cost and still meets the campaign's deadline. *Define acceptance criteria before selecting a more complex model.*

Reading anchor: Bucaioni, published Table 1, pp. 6–7 · Bass, Ch. 14 · Kalinowski, §13.4

48 / 55 Robustness: invalid, unfamiliar, or ordinary?

Block 2 / Quality concerns • TBD



CUES

- Presenter: **TBD**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

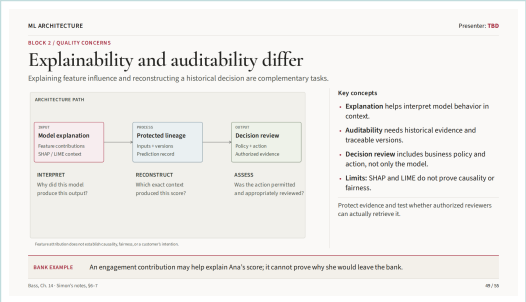
TO SAY

Different inputs need different responses; do not treat every unusual value as the same failure. **Contract validity** and model applicability are distinct. **OOD inputs** may be structurally valid but poorly represented in training. **Confidence** alone is not a guaranteed detector of unfamiliar inputs. **Fallbacks** must be validated for the intended situation. *At our bank,* a missing activity value is invalid; a new customer population may be valid but outside the model's reliable coverage. *An architecture needs an explicit response when the model should not be trusted.*

Reading anchor: Bucaioni, published Table 1, pp. 6–7 · Bass, Ch. 14 · Simon's notes, §2–4

49 / 55 Explainability and auditability differ

Block 2 / Quality concerns • TBD



CUES

- Presenter: **TBD**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

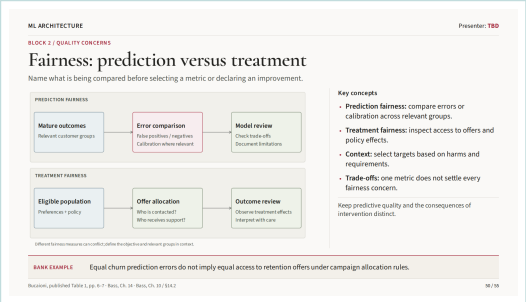
TO SAY

Explaining feature influence and reconstructing a historical decision are complementary tasks. **Explanation** helps interpret model behavior in context. **Auditability** needs historical evidence and traceable versions. **Decision review** includes business policy and action, not only the model. **Limits**: SHAP and LIME do not prove causality or fairness. *At our bank,* an engagement contribution may help explain Ana's score; it cannot prove why she would leave the bank. *Protect evidence and test whether authorized reviewers can actually retrieve it.*

Reading anchor: Bass, Ch. 14 · Simon's notes, §6–7

50 / 55 Fairness: prediction versus treatment

Block 2 / Quality concerns • TBD



CUES

- Presenter: **TBD**.
- Animated build: click 3 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

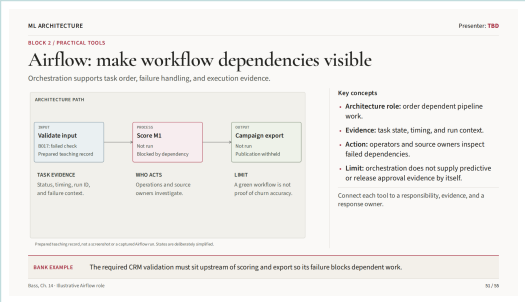
TO SAY

Name what is being compared before selecting a metric or declaring an improvement. **Prediction fairness**: compare errors or calibration across relevant groups. **Treatment fairness**: inspect access to offers and policy effects. **Context**: select targets based on harms and requirements. **Trade-offs**: one metric does not settle every fairness concern. *At our bank,* equal churn prediction errors do not imply equal access to retention offers under campaign allocation rules. *Keep predictive quality and the consequences of intervention distinct.*

Reading anchor: Bucaioni, published Table 1, pp. 6–7 · Bass, Ch. 14 · Bass, Ch. 10 / §14.2

51 / 55 Airflow: make workflow dependencies visible

Block 2 / Practical tools • TBD



CUES

- New block: **Block 2 / Practical tools**.
- Presenter: **TBD**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

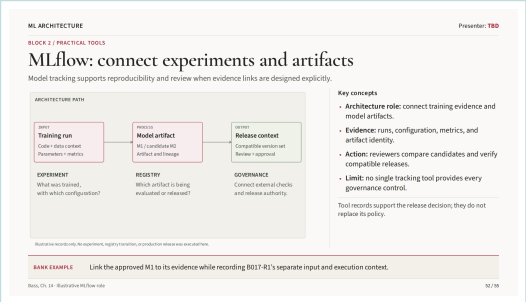
TO SAY

Orchestration supports task order, failure handling, and execution evidence. **Architecture role**: order dependent pipeline work. **Evidence**: task state, timing, and run context. **Action**: operators and source owners inspect failed dependencies. **Limit**: orchestration does not supply predictive or release approval evidence by itself. *At our bank,* the required CRM validation must sit upstream of scoring and export so its failure blocks dependent work. *Connect each tool to a responsibility, evidence, and a response owner.*

Reading anchor: Bass, Ch. 14 · Illustrative Airflow role

52 / 55 MLflow: connect experiments and artifacts

Block 2 / Practical tools • TBD



CUES

- Presenter: **TBD**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

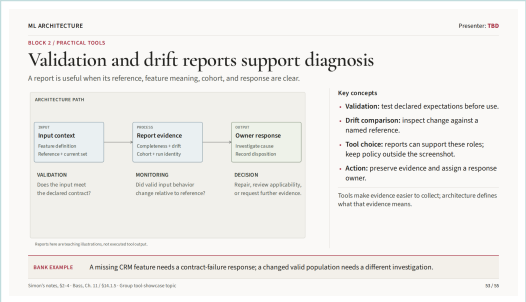
TO SAY

Model tracking supports reproducibility and review when evidence links are designed explicitly. **Architecture role**: connect training evidence and model artifacts. **Evidence**: runs, configuration, metrics, and artifact identity. **Action**: reviewers compare candidates and verify compatible releases. **Limit**: no single tracking tool provides every governance control. *At our bank,* link the approved M1 to its evidence while recording B017-R1's separate input and execution context. *Tool records support the release decision; they do not replace its policy.*

Reading anchor: Bass, Ch. 14 · Illustrative MLflow role

53 / 55 Validation and drift reports support diagnosis

Block 2 / Practical tools • TBD



CUES

- Presenter: **TBD**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

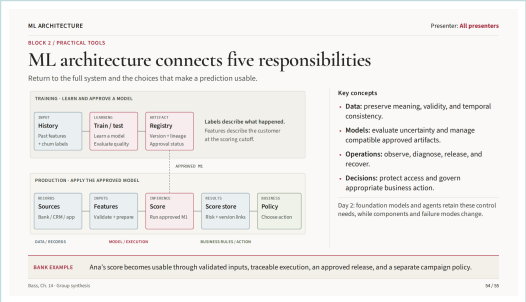
TO SAY

A report is useful when its reference, feature meaning, cohort, and response are clear. **Validation**: test declared expectations before use. **Drift comparison**: inspect change against a named reference. **Tool choice**: reports can support these roles; keep policy outside the screenshot. **Action**: preserve evidence and assign a response owner. *At our bank,* a missing CRM feature needs a contract-failure response; a changed valid population needs a different investigation. *Tools make evidence easier to collect; architecture defines what that evidence means.*

Reading anchor: Simon’s notes, §2–4 · Bass, Ch. 11 / §14.1.5 · Group tool-showcase topic

54 / 55 ML architecture connects five responsibilities

Block 2 / Practical tools • All presenters



CUES

- Presenter: **All presenters**.
- Animated build: click 4 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

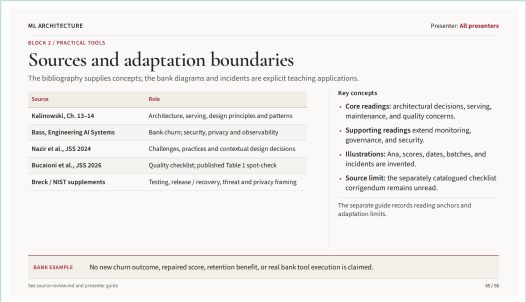
TO SAY

Return to the full system and the choices that make a prediction usable. **Data**: preserve meaning, validity, and temporal consistency. **Models**: evaluate uncertainty and manage compatible approved artifacts. **Operations**: observe, diagnose, release, and recover. **Decisions**: protect access and govern appropriate business action. *At our bank,* ana's score becomes usable through validated inputs, traceable execution, an approved release, and a separate campaign policy. *Day 2: foundation models and agents retain these control needs, while components and failure modes change.*

Reading anchor: Bass, Ch. 14 · Group synthesis

55 / 55 Sources and adaptation boundaries

Block 2 / Practical tools • All presenters



CUES

- Presenter: **All presenters**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

TO SAY

The bibliography supplies concepts; the bank diagrams and incidents are explicit teaching applications. **Core readings**: architectural decisions, serving, maintenance, and quality concerns. **Supporting readings** extend monitoring, governance, and security. **Illustrations**: Ana, scores, dates, batches, and incidents are invented. **Source limit**: the separately catalogued checklist corrigendum remains unread. *At our bank,* no new churn outcome, repaired score, retention benefit, or real bank tool execution is claimed. *The separate guide records reading anchors and adaptation limits.*

Reading anchor: See source-review.md and presenter guide